Final Project

Computer Programming I

I. Overview

Students will identify a real-world business process, manual operation, or organizational workflow that can be improved or automated using a C++ system. Each project should focus on a **specific stakeholder or beneficiary business**, demonstrating innovation, efficiency, and proper use of programming concepts such as data structures, functions, file handling, and object-oriented programming.

II. Objective

The goal of this final project is to apply your C++ programming knowledge to develop a functional system that automates or transforms a manual business process into a computer-based application. By working with a real stakeholder or beneficiary business, students will gain practical experience in understanding user needs, problem-solving, and delivering an effective solution.

III. Project Requirements

Your project must:

a. Be written entirely in C++.

b. Demonstrate the use of the following programming concepts:

- **i.** **Variables** and **control structures** (if, loops, switch)
- **ii.** **Functions** for modular design
- **iii.** **Arrays** or other **data structures** (e.g., struct, class, vector)
- **iv.** Implement either of the following methods for saving and managing data:
  - **1.** File Handling (text or binary files)
  - **2.** Database Integration (e.g., MySQL or PostgreSQL) using C++ connectors
- **v.** Apply **Object-Oriented Programming (OOP)** principles such as classes and objects
- **vi.** Include a **user-friendly interface** (menu-driven)
- **vii.** Allow users to perform basic **CRUD** operations (Create, Read, Update, Delete) where applicable
- **viii.** Be **original, functional, and well-documented** (include comments, clear variable names, and explanations)
- **ix.** Identify and describe the **stakeholder or beneficiary business**, including:
  - **1.** Who will use the system?
  - **2.** What problems it solves for them?
  - **3.** How it improves their process or workflow?

IV. Project Deliverables

a. Each group must submit:

- **i.** **Source Code (.cpp files)** – Complete, tested, and properly organized.
- **ii.** **Documentation (Word/PDF)** that includes:
  - **1.** **Project Title and Group Members** – The name of your system/project and the list of all group members.
  - **2.** **Project Description** – A brief overview of what the system does and the problem it solves.
  - **3.** **Project Objectives** – Clear goals your system aims to achieve.

4. **SDG Alignment** – How your project supports one or more Sustainable Development Goals.
5. **Stakeholder/Beneficiary Business Description** – Who will use the system and how it benefits them.
6. **Functional Requirements** – Key features and actions your system can perform.
7. **System Flowchart** – Visual representation of the system's workflow or logic.
8. **Technology Transfer / Implementation –** *(Optional but encouraged)* Explain how the system could be deployed, applied, or transferred to the stakeholder/beneficiary, including usability considerations, step-by-step usage instructions, and possible real-world application.

iii. **Demonstration/Presentation** – Group members will present and explain:

1. Their system, its features, and implementation details
2. How it addresses the needs of the identified stakeholder or beneficiary business**.**

V. Criteria

Criteria	Description	Points
1. System Functionality	Evaluates correctness, completeness, and usability of the C++ system. Includes proper use of control structures, functions, data structures, file handling, OOP, and menu-driven interface. • Fully functional and accurate output • Proper application of programming concepts • Logical flow and error handling	**50 pts**
2. Documentation	Assesses quality, completeness, relevance, and accuracy of the written report. Must include all required sections: Title, Description, Objectives, SDG Alignment, Stakeholder/Beneficiary, Functional Requirements, Flowchart, and (optional) Technology Transfer. • Content accuracy and relevance to the developed system • Completeness and logical organization • Clarity, coherence, and grammar	**30 pts**
3. Presentation & Communication	Measures clarity and effectiveness of group presentation, demonstration of the system, and ability to answer questions confidently.	**15 pts**
4. Peer Evaluation	Focuses on each member's **initiative, collaboration, reliability, and problem-solving** during project development. Based on peer feedback and instructor observation.	**5 pts**
5. Bonus: Technology Transfer / Implementation	Stakeholder/beneficiary. Bonus is awarded if the **system is designed in a way that the client or stakeholder can realistically utilize it** in their workflow.	**+5 pts (Bonus)**